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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Time: 03 Hours

Code: 102603 ***Subject: Manufacturing Technology*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) Which of the following process is not grouped under metal removal process?
(i) boring (iii) tumbling
(ii) milling (iv) rolling

(b) Machine tools can be classified as
(i) geometric shape of the work piece (iii) function of the machine
(ii) number of the work piece (iv) all of the mentioned

(c) Structural sections such as rails, angles, I-beams are made by
(i) Hot rolling (iii) Hot piercing
(ii) Hot drawing (iv) Hot extrusion

(d) Ultrasonic machining is best suited for
(i) Non-ferrous metal (iii) Amorphous solids
(ii) Brittle materials (iv) Dielectrics

(e) Which of the following processes has the highest metal removal rate?
(i) Chemical machining (iii) Ultrasonic machining
(ii) Abrasive machining (iv) Electron beam machining

(f) In abrasive jet machining metal removal takes place due to
(i) evaporation (iii) corrosion
(ii) erosion (iv) melting

(g) Taylor's principle is concerned with
(i) Pneumatic comparators (iii) Gauging measurements
(ii) Interferometric measurements (iv) Angular measurement

(h) Limit Gauge
(i) checks whether the part is made within the specified tolerance
(ii) measures both upper and lower specification limit
(iii) measures the value of the lower specification limit
(iv) measures the value of the upper specification limit

(i) Tachometers are used to measure:
(i) Acceleration (iii) Time
(ii) Angular velocity (iv) Force

(j) Tolerance of gauges can be checked by
(i) optical flat (iii) auto collimator
(ii) sine bar (iv) spirit level

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Q.2 Solve both questions : **[7+7=14]**

(a) Differentiate Hot and Cold Working Process.

(b) Discuss the classification of press tools.

Q.3 Solve both questions : **[7+7=14]**

(a) Describe briefly the design principles for jigs and fixtures.

(b) Describe briefly the accessories of jigs and fixtures.

Q.4 Solve both questions : **[7+7=14]**

(a) State and explain the Taylor's principle of gauge design with neat sketch of Plug gauge and Snap gauges.

(b) Explain with the help of a diagram the principle of working of a sine bar for angular measurement. List the advantages and limitations of sine bar.

Q.5 Solve both questions : **[7+7=14]**

(a) Why is it necessary to give a tolerance on engineering dimension? Give an example of both unilateral and bilateral tolerances.

(b) Explain the classification of material handling equipments.

Q.6 Solve both questions : **[7+7=14]**

(a) Explain Abrasive Jet Machining (AJM) with neat sketch. Also explain its advantages and disadvantages.

(b) Explain Water Jet Machining (WJM) with neat sketch. Also explain its advantages and disadvantages.

Q.7 Solve both questions : **[7+7=14]**

(a) Explain the basic principle of Electro Discharge Machining Die Sinking EDM explain different process parameters (Input parameters) used.

(b) Enumerate the difference between AJM and WJM.

Q.8 Solve both questions : **[7+7=14]**

(a) Explain Ultrasonic Machining with neat sketch. Also mention different process parameters.

(b) Explain Electro Chemical Machining with neat sketch. Also explain its advantages and disadvantages.

Q.9 Solve both questions : **[7+7=14]**

(a) Enumerate the difference between Laser Beam Machining (LBM) and Electron beam Machining (EBM).

(b) Explain wire-cut EDM with neat sketch. Also explain its advantages and disadvantages.

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